

Genetic report for longevity

Skin and Skin Care

Barriers, sun and pigmentation, skin cancer, photoaging, acne, seborrheic dermatitis and topical assets

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The skin is the organ where aging is first seen and where more can be done on the outside. This report reads the genes of skin barrier, sun response and pigmentation, the risk of skin cancer, wrinkles and spots, acne, seborrheic dermatitis and antioxidant defense and collagen, and ends with the topical assets and routine that genetics favors.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

46 variants	37 in the chip	8 imputed (DR2≥0.8)	1 not found	16 Scores with percentile
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Overall reading. The genetics of the skin is, in general, favourable. The response to the sun is of the best: no strong variant of MC1R, IRF4 without the T allele, 1 allele "dark" in SLC45A2 and TYR 402R/R; the scores make it easy to tan, the few burns in childhood and the few freckles on the good side (15, 1 and 6 percentiles) and the skin genetically darker than 94% of the reference population. Skin cancer also comes out well: squamous cell at the 3rd percentile, basal cell in the 20 and the two public melanoma scores at the 7 and 56 percentiles. These two do not confirm the 89th percentile of a previous report (ownership method); as the number of nevi (3 loci score, the 88th percentile) and the CDKN2A/MTAP region pull up, annual dermoscopy continues to make sense. Wrinkles (percentile 0.3; allele count score, weak evidence) and skin aging (the 5th percentile, few loci) are favourable, and genetic glycation is low (HbA1c percentile 7). In seborrheic dermatitis, the GWAS score is low (the 9th percentile); CARD9 TT (stronger response to fungi) and TLR2 P631H heterozygous are the only signs against, both of weak evidence. In the barrier, rare mutations of FLG are absent; R501X has not been read (an earlier report gave GG) and rs542799026 remains "possible, unconfirmed". Isotretinoin weighs more on the skin (dryness, photosensitivity) than any gene in this report.

Favourable

- **Tanning difficulty (burning instead of tanning):** the 16th percentile
- **Solar burns in childhood:** the 1st percentile
- **Skin color (darker):** the 94th percentile
- **Melanoma (24 significant genomic variants):** the 6th percentile
- **Basal cell carcinoma:** the 20th percentile
- **Squamous cell carcinoma:** the 3rd percentile
- **Genetic glycosylated hemoglobin (HbA1c):** the 7th percentile
- **Sardines:** the 6th percentile
- **Face wrinkles:** the 0th percentile
- **Skin aging (microtopography):** the 5th percentile
- **Seborrheic dermatitis:** the 9th percentile
- **FLG rs138726443 GG:** R2447X: rare filaggrin mutation.
- **FLG rs150597413 GG:** S3247X: rare filaggrin mutation.
- **MC1R rs1805007 CC:** R151C: "R" variant (strong) of MC1R; red hair, freckles, burns and ~2× risk of melanoma per copy.
- **MC1R rs1805008 CC:** R160W: variant "R" (strong) of MC1R.
- **MC1R rs1805009 GG:** D294H: variant "R" (strong) of MC1R.
- **MC1R rs1805006 CC:** D84E: variant "R" (strong), rare.
- **MC1R rs11547464 GG:** R142H: variant "R" (strong), rare.
- **MC1R rs1110400 TT:** I155T: "r" variant (weak) of MC1R.
- **SLC45A2 rs16891982 CG:** L374F: the G allele (374F) clears skin and hair and increases the risk of melanoma; the C allele (374L) is the "dark".
- **TYR rs1126809 GG:** R402Q: Allele A (402Q) gives a thermosensitive tyrosinase, less melanin and more risk of melanoma (but less vitiligo).
- **CDKN2A rs3731249 CC:** A148T: moderate risk variant of CDKN2A (p16), ~1.5–2× risk of melanoma per copy.
- **MITF rs149617956 GG:** E318K: rare variant of MITF that increases ~2–3× the risk of melanoma and the number of nevi.
- **CLEC7A rs16910526 AA:** Y238X (Dectin-1): the rare allele truncates the β-glucans receptor and is associated with mucocutaneous and nail infections.
- **ASIP rs4911412 GG:** ASIP (agouti protein): favors pheomelanin; pigmentation locus and melanoma.
- **CASP8 rs10931936 CC:** CASP8: apoptosis; melanoma locus.

Unfavourable

- **Number of nevi (signs):** the 88th percentile
- **Melanoma (previous report, proprietary method):** the 89th percentile (\$10 profile)
- **FLG rs542799026 CG:** Rare non-sensory variant of FLG, reads in a previous analysis as "possible" and unconfirmed.
- **CARD9 rs4077515 TT:** S12N: T allele (12N) is associated with stronger CARD9 signaling (more IL-17 against fungi) and Crohn's disease; hypothesis: stronger inflammatory response to Malassezia in seborrheic dermatitis.
- **MTAP rs869330 AA:** CDKN2A/MTAP region (9p21): the strongest common locus of melanoma and the number of nevi.
- **PARP1 rs2695237 TT:** PARP1: repair of DNA; melanoma locus.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 37 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Skin barrier and hydration

Filaggrin (FLG) is the protein that gives rise to the "natural hydration factor" of the stratum corneum; the null mutations of FLG (about 8–10% of people have one) dry the skin and triplicate the risk of eczema. Aquaporin 3 (AQP3) carries water and glycerol in the epidermis. Note: the list of objectives does not have its own mechanism for the skin barrier; the closest barrier is used.

Ageing mechanisms: Intestinal Barrier dysfunction; Chronic Inflammation; Remodelling of the extracellular matrix

Favourable

- FLG rs138726443 GG
- FLG rs150597413 GG

Unfavourable

- FLG rs542799026 CG

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG	rs138726443 1:152307547 · G/A/C/T	GG	chip	A (0; 0%)	strong	R2447X: rare filaggrin mutation.
FLG	rs150597413 1:152305146 · G/A/C/T	GG	chip	T (0; 0%)	strong	S3247X: rare filaggrin mutation.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG	rs542799026 1:152306769 · G/A/C	CG	chip	C (1; 0%)	moderate	Rare non-sensory variant of FLG, reads in a previous analysis as "possible" and unconfirmed. Allele C absent in the reference population (frequency 0%): a so-called isolated heterozygous variant in the chip is often artefact. Not confirmed; confirm by sequencing only if there is eczema or very dry skin.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG	rs558269137 1:152312601 · ACTGACTG/ACTG	AT	chip	—	strong	2282del4: second most common null mutation; it is a base 4 deletion. The chip uses probes for this deletion and the reading is ambiguous (non-interpretable).
AQP3	rs17553719 9:33447581 · T/A/C/G	TT	chip	—	weak	Aquaporin 3 promoter: used in commercial "hydration" tests; weak evidence. Information.
SPINK5	rs2303067 5:148101392 · A/C/G/T	AG	chip	—	weak	E420K (LEKTI, desquamation protease inhibitor): weak associations with atopy. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG	rs61816761 1:152313385 · G/A/C/T	—	SNP not found in the data (Imputed with DR2=0.48 < 0.8)	A (—; 1%)	strong	R501X: allele A (strand +; T in the gene) creates a stop codon and is the most common filaggrin mutation in the reference population; 1 copy already reduces filaggrin and increases dryness and eczema. It is not on the chip and the imputation does not have enough quality; a

previous genetic report indicated GG (non-carrier).

Sun, tanning and pigmentation

MC1R, IRF4, SLC45A2, SLC24A5, TYR, OCA2/HERC2 and ASIP decide the amount and type of melanin (brown eumelanin, which protects, or red pheomelanin, which protects little and generates radicals). Easiness scores of tanning, sunburn in childhood, skin color and freckles.

Ageing mechanisms: DNA Damage and DNA Repair Deficit; Oxidative stress; Somatic Mutation Accumulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Tanning difficulty (burning instead of tanning)	Favourable	the 16th percentile · z=-0.98	Higher score : less bronze and more burning. PGS001937 (PGS Catalog; Penalized regression (bigstatsr); Ease of skin tanning; PGP000263, PMID 34995502). Variants with genotype: 15724 of 16197 (covering of 97.9% coefficients).
Solar burns in childhood	Favourable	the 1st percentile · z=-2.23	Higher score = more burns. PGS001257 (PGS Catalog; snpnet; Childhood sunburn; PGP000244, PMID 35324888). Variants with genotype: 444 of 519 (covering of 93.8% coefficients).
Skin color (darker)	Favourable	the 94th percentile · z=1.66	Higher score : darker skin within the reference population (more natural photoprotection). PGS001897 (PGS Catalog; Penalized regression (bigstatsr); Skin color; PGP000263, PMID 34995502). Variants with genotype: 15325 of 15817 (covering of 97.7% coefficients).
Sardines	Favourable	the 6th percentile · z=-1.55	Higher score = more freckles. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 10 of 11 loci with genotype. Studies: Endo C (Freckles); Eriksson N (Freckling); Farre X (Freckles); Sulem P (Freckles). Variants with genotype: 10 of 11 (covering of 90.9% coefficients).

Favourable

- Tanning difficulty (burning instead of tanning) (the 16th percentile)
- Solar burns in childhood (the 1st percentile)
- Skin color (darker) (the 94th percentile)
- Freckles (the 6th percentile)
- MC1R rs1805007 CC
- MC1R rs1805008 CC
- MC1R rs1805009 GG
- MC1R rs1805006 CC
- MC1R rs11547464 GG
- MC1R rs1110400 TT
- SLC45A2 rs16891982 CG
- TYR rs1126809 GG

Unfavourable

None.

Favourable variants (8)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MC1R	rs1805007 16:89919709 · C/A/G/T	CC	chip	T (0; 7%)	strong	R151C: "R" variant (strong) of MC1R; red hair, freckles, burns and ~2× risk of melanoma per copy.
MC1R	rs1805008 16:89919736 · C/T	CC	chip	T (0; 6%)	strong	R160W: strong variant of MC1R.
MC1R	rs1805009 16:89920138 · G/A/C	GG	chip	C (0; 1%)	strong	D294H: strong variant of MC1R.
MC1R		CC	chip	A (0; 1%)	strong	D84E: variant "R" (strong), rare.

	rs1805006 16:89919510 · C/A/G					
MC1R	rs11547464 16:89919683 · G/A	GG	chip	A (0; 1%)	strong	R142H: variant "R" (strong), rare.
MC1R	rs1110400 16:89919722 · T/C	TT	chip	C (0; 1%)	moderate	I155T: "r" variant (weak) of MC1R.
SLC45A2	rs16891982 5:33951588 · C/A/G	CG	chip	G (1; 94%)	strong	L374F: the G allele (374F) clears skin and hair and increases the risk of melanoma; the C allele (374L) is the "dark".
TYR	rs1126809 11:89284793 · G/A	GG	imputation DR2=1.00	A (0; 25%)	moderate	R402Q: Allele A (402Q) gives a thermosensitive tyrosinase, less melanin and more risk of melanoma (but less vitiligo).

Typical / intermediate variants (6)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MC1R	rs2228479 16:89919532 · G/A/C	GG	chip	A (0; 7%)	moderate	V92M: "r" variant (weak) of MC1R; small effect on freckles and melanoma.
IRF4	rs12203592 6:396321 · C/ G/T	CC	chip	T (0; 12%)	strong	Intronic (tyrosinase enhancer): the T allele gives more sensitive skin to the sun, freckles, sunspots and white hair earlier.
SLC24A5	rs1426654 15:48134287 · A/G/T	AA	chip	A (2; 100%)	strong	A111T: allele A (111T) clears the skin; it is almost fixed in the reference population.
TYR	rs1042602 11:89178528 · C/A	CA	chip	A (1; 37%)	weak	S192Y: Allele A (192Y) is associated with lighter skin and freckles.
HERC2	rs12913832 15:28120472 · A/C/G	AG	chip	G (1; 64%)	strong	OCA2 enhancer: G allele turns off OCA2 in iris melanocytes (blue eyes) and clears the skin a little; it is associated with a higher risk of melanoma.
OCA2	rs1800407 15:27985172 · C/T	CC	chip	T (0; 8%)	moderate	R419Q: the T allele (419Q) clears the eyes (green / hazelnut) and increases a little the risk of melanoma.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MC1R	rs885479 16:89919746 · G/A/C	GG	chip	—	weak	R163Q: very weak variant, almost without effect on pigmentation in the reference population. Information.

Skin cancer

Melanoma, basal cell carcinoma and squamous cell. In addition to the pigmentation genes, CDKN2A/MTAP, TERT, MITF, PLA2G6, PARP1 and the number of nevi (signals) enter.

Ageing mechanisms: Genomic instability; Accumulation of Somatic Mutations DNA Damage and DNA Repair Deficit; Somatic / Tissue Mosaicism

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Melanoma (24 significant genomic variants)	Favourable	the 6th percentile · z=-1.43	Higher score : higher risk. PGS000079 (PGS Catalog; Genome-wide signifiers; Melanoma; PGP000050, PMID 33579919). Variants with genotype: 22 of 24 (covering of 94.2% coefficients).
Melanoma (505 variants, lassosum)	Typical / intermediate	the 56th percentile · z=-0.13	Higher score : higher risk. PGS000415 (PGS Catalog; lassosum; Melanomas of skin (diagnosis or history); PGP000118, PMID 32991828). Variants with genotype: 438 of 505 (covering of 84.1% coefficients).
Basal cell carcinoma	Favourable	the 20th percentile · z=-0.88	Higher score : higher risk. PGS000448 (PGS Catalog; lassosum; Basal cell carcinoma; PGP000118, PMID 32991828). Variants with genotype: 173 of 183 (covering of 95.5% coefficients).
Squamous cell carcinoma	Favourable	the 3rd percentile · z=-1.75	Higher score : higher risk. PGS000464 (PGS Catalog; Pruning and Thresholding (P+T); Squamous cell carcinoma; PGP000118, PMID 32991828). Variants with genotype: 13 of 14 (covering of 95.8% coefficients).
Number of nevi (signs)	Unfavourable	the 88th percentile · z=1.30	Higher score = more nevus (the highest risk factor of melanoma). Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 3 of 4 loci with genotype. Studies: Chen Y (Nevus count); Falchi M (Cutaneous nevi). Variants with genotype: 3 of 4 (covering of 75.0% coefficients).
Melanoma (previous report, proprietary method)	Unfavourable	the 89th percentile	Result of a previous report with another method; the two public scores of this report do not confirm it (see reading). Source: \$10 profile.

Favourable

- Melanoma (24 significant genomic variants) (the 6th percentile)
- Basal cell carcinoma (the 20th percentile)
- Squamous cell carcinoma (the 3rd percentile)
- CDKN2A rs3731249 CC
- MITF rs149617956 GG
- ASIP rs4911412 GG
- CASP8 rs10931936 CC

Unfavourable

- Number of nevi (signals) (the 88th percentile)
- Melanoma (previous report, proprietary method) (the 89th percentile)
- MTAP rs869330 AA
- PARP1 rs2695237 TT

Favourable variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CDKN2A	rs3731249 9:21970917 · C/A/G/T	CC	chip	T (0; 2%)	strong	A148T: moderate risk variant of CDKN2A (p16), ~1.5-2× risk of melanoma per copy.
MITF	rs149617956 3:69964940 · G/A/T	GG	chip	A (0; 0%)	strong	E318K: rare variant of MITF that increases ~2-3× the risk of melanoma and the number of nevi.
ASIP	rs4911412 20:34128629 · A/C/G/T	GG	imputation DR2=1.00	A (0; 30%)	moderate	ASIP (agouti protein): favors pheomelanin; pigmentation locus and melanoma. Meaning: GWAS Catalog: Cutaneous

melanoma (MTAG) (Liyanage U 2021; p=2e-38); Allele A ↑

CASP8	rs10931936 2:201279205 · T/A/C/G	CC	imputation DR2=0.99	T (0; 28%)	moderate	CASP8: apoptosis; melanoma locus. Meaning: GWAS Catalog: Cutaneous malignant melanoma (Landi MT 2020; p=2e-12); T ↑ allele
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Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MTAP	rs869330 9:21804618 · A/C/G	AA	imputation DR2=1.00	A (2; 50%)	moderate	CDKN2A/MTAP region (9p21): the strongest common locus of melanoma and the number of nevi. Meaning: GWAS Catalog: Invasive melanoma (Ingold N 2024; p=3e-35); Allele A ↑
PARP1	rs2695237 1:226415934 · T/A/C/G	TT	imputation DR2=0.99	T (2; 66%)	moderate	PARP1: repair of DNA; melanoma locus. Meaning: GWAS Catalog: Cutaneous melanoma (MTAG) (Liyanage U 2021; p=4e-21); T ↑ allele

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TERT	rs7705526 5:1285859 · C/ A/G/T	CA	chip	A (1; 34%)	moderate	TERT: telomerase; alleles that extend the telomeres increase the risk of melanoma. Meaning: GWAS Catalog: Cutaneous malignant melanoma (Landi MT 2020; p=2e-14); Allele A ↑
PLA2G6	rs132941 22:38149935 · T/A/C/G	CT	imputation DR2=0.99	T (1; 55%)	moderate	PLA2G6: melanoma locus and nevi. Meaning: GWAS Catalog: Cutaneous melanoma (MTAG) (Liyanage U 2021; p=3e-34); T ↑ allele
MX2	rs408825 21:41371569 · C/T	CT	imputation DR2=1.00	T (1; 62%)	moderate	MX2: melanoma locus. Meaning: GWAS Catalog: Cutaneous malignant melanoma (Landi MT 2020; p=1e-32); T ↑ allele

Photoaging, wrinkles and stains

The wrinkles and flaccidity come mainly from UV radiation (degradation of collagen by MMP-1, solar elastosis) and glycation. Genes of wrinkles (STXBP5L, MMP1/MMP3), spots and freckles (IRF4, BNC2) and scores of facial aging and seborrheic keratosis.

Ageing mechanisms: Remodelling of the extracellular matrix; Fragmentation of Elastin; Cell Senescence; Oxidative stress

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Facial aging ("seems older than it is")	Typical / intermediate	the 29th percentile · z=-0.64	Higher score = more likely to look older. PGS001072 (PGS Catalog; snpnet; Facial aging, looking 'older than you are'; PGP000244, PMID 35324888). Variants with genotype: 27 of 39 (covering of 75.7% coefficients).
Seborrheic keratoses	Typical / intermediate	the 40th percentile · z=-0.21	Higher score = more keratoses (brown benign lesions typical of aging). PGS001140 (PGS Catalog; snpnet; Seborrheic keratosis (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 1166 of 1367 (covering of 87.4% coefficients).
Face wrinkles	Favourable	the 0th percentile · z=-2.43	Higher score = more wrinkles. Few loci; low reliability. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 28 of 34 loci with genotype. Studies: Chen Y (Facial wrinkles) Chen Y (Facial wrinkles) Chen Y (Facial wrinkles) Chen Y (Facial wrinkles) Hamer MA (Facial wrinkles); Lee SG (Facial wrinkles) Lee SG (Facial wrinkles (under eye)); Okuno R (Wrinkles in other corners of eyes) Variants with genotype: 28 of 34 (covering of 82.4% coefficients).
Skin aging (microtopography)	Favourable	the 5th percentile · z=-1.88	Higher score = more measured aging. Few loci; low reliability. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 3 of 4 loci with genotype. Studies: Law MH (Skin aging) Variants with genotype: 3 of 4 (covering of 77.8% coefficients).

Favourable

- Facial wrinkles (the 0th percentile)
- Skin aging (microtopography) (the 5th percentile)

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BNC2	rs16935073 9:16795792 · A/C/G	AA	chip	C (0; 6%)	moderate	BNC2: locus of freckles and facial pigments. Meaning: GWAS Catalog: Facial pigmentation measurement (UV light) (Shin JG 2020; p=2e-46); allele C ↑

Acne and sebaceous gland

Acne loci score (GWAS of 2022 with 46 loci: WNT10A, OVOL1, FST, TGFB2 and others, especially in the shape and development of the hair follicle). The isotretinoin already acts at the main point (sebaceous gland).

Ageing mechanisms: Chronic Inflammation; Endocrine dysregulation; Dysbiosis

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Acne	Typical / intermediate	the 40th percentile · z=-0.24	Higher score : higher risk of acne. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 48 of 52 loci with genotype. Studies: He L (Acne (severe)); Mitchell BL (acne vulgaris); Petridis C (Acne (severe)); Teder-Laving M (acne vulgaris); Verma A (Acne (PheCode 706.1)). Variants with genotype: 48 of 52 (covering of 92.3% coefficients).

Favourable

None.

Unfavourable

None.

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
WNT10A	rs74333950 · 2:218881570 · T/G	TT	chip	T (2; 88%)	moderate	WNT10A: development of hair follicle; acne locus. Meaning: GWAS Catalog: acne vulgaris (Mitchell BL 2022; p=3e-23); allele T ↑
FST	rs629725 · 5:53335237 · T/A/C	CT	imputation · DR2=0.98	T (1; 31%)	moderate	Follistatin (FST): severe acne locus. Meaning: GWAS Catalog: acne vulgaris (Mitchell BL 2022; p=4e-28); allele T ↑
SEMA4B	rs34560261 · 15:90191194 · C/T	CC	chip	C (2; 86%)	moderate	SEMA4B: severe acne locus. Meaning: GWAS Catalog: acne vulgaris (Mitchell BL 2022; p=3e-35); allele C ↑

Seborrheic dermatitis

Seborrheic dermatitis is an inflammatory reaction to *Malassezia* yeast in areas rich in tallow. Fungal response genes (CARD9, Dectin-1/CLEC7A, TLR2) and the loci score of GWAS of seborrheic dermatitis.

Ageing mechanisms: Chronic Inflammation; Dysbiosis; Immunosenescence

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Seborrheic dermatitis	Favourable	the 9th percentile · z=-1.31	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 23 of 24 loci with genotype. Studies: Verma A (Seborrheic dermatitis (PheCode 690.1)). Variants with genotype: 23 of 24 (covering of 93.9% coefficients).

Favourable

- Seborrheic dermatitis (the 9th percentile)
- CLEC7A rs16910526 AA
- TLR2 rs5743708 GG

Unfavourable

- CARD9 rs4077515 TT
- TLR2 rs5743704 CA

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CLEC7A	rs16910526 · 12:10118488 · A/C/G	AA	chip	C (0; 7%)	moderate	Y238X (Dectin-1): the rare allele truncates the β-glucans receptor and is associated with mucocutaneous and nail infections.
TLR2	rs5743708 · 4:153705165 · G/A	GG	chip	A (0; 2%)	weak	R753Q: the rare allele reduces TLR2 signaling; it is associated with severe atopic dermatitis and staphylococci infections.

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CARD9	rs4077515 · 9:136372044 · C/A/T	TT	chip	T (2; 40%)	weak	S12N: T allele (12N) is associated with stronger CARD9 signaling (more IL-17 against fungi) and Crohn's disease; hypothesis: stronger inflammatory response to <i>Malassezia</i> in seborrheic dermatitis.
TLR2	rs5743704 · 4:153704799 · C/A/T	CA	chip	A (1; 3%)	weak	P631H: the rare allele reduces a little the TLR2 signaling. Small effect.

Collagen, antioxidants and skin glycation

Antioxidant enzymes (SOD2, catalase, GPX1, NQO1, GSTP1), type I collagen and AGE/RAGE axis, which hardens the collagen of the dermis and gives it yellowish color.

Ageing mechanisms: Oxidative stress; Redox imbalance; Accumulation of Advanced Glycation Final Products (AGEs); Remodelling of the extracellular matrix

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Genetic glycated hemoglobin (HbA1c)	Favourable	the 7th percentile · z=-1.51	Higher score : more glycation throughout life (also of the collagen of the dermis). PGS001953 (PGS Catalog; Penalized regression (bigstatsr); Glycated haemoglobin (HbA1c); PGP000263, PMID 34995502). Variants with genotype: 44856 of 46566 (covering of 96.2% coefficients).

Favourable

- Genetic glycated hemoglobin (HbA1c) (the 7th percentile)

Unfavourable

None.

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL1A1	rs1800012 · 17:50200388 · C/A	CC	chip	A (0; 19%)	moderate	Sp1: the allele "s" (A in strand +) alters the proportion of collagen type I chains.
GPX1	rs1050450 · 3:49357401 · G/A	GA	chip	A (1; 34%)	weak	P198L: Allele A (198L) gives less activity of glutathione peroxidase 1.
NQO1	rs1800566 · 16:69711242 · G/A	GG	chip	A (0; 21%)	moderate	P187S: Allele A (187S) gives an unstable NQO1; in AA the enzyme almost does not exist (less defense against quinones and oxidative stress).
AGER	rs2070600 · 6:32183666 · C/T	CC	chip	T (0; 5%)	moderate	G82S: the T allele lowers sRAGE (AGEs bait) and increases inflammatory signaling of RAGE.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SOD2	rs4880 · 6:159692840 · A/G	GG	chip	—	weak	V16A: the Ala allele (G at strand +) enters better in mitochondria and gives more activity (plus H2O2); Val/Val gives less. Effects in opposite directions according to the context. Information.
CAT	rs1001179 · 11:34438684 · C/G/T	CC	chip	—	weak	-262C>T (catalase promoter): effects on the contradictory activity between studies. Information.
GSTP1	rs1695 · 11:67585218 · A/G/T	AG	chip	—	weak	I105V: changes the specificity of GSTP1 substrate (more active for some, less for others). Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Somatic Mutation Accumulation	Number of nevi (signals) (the 88th percentile); Melanoma (previous report, proprietary method) (the 89th percentile); MTAP AA; PARP1 TT	Tanning difficulty (burning instead of tanning) (the 16th percentile); Solar burns in childhood (the 1st percentile); Skin color (darker) (the 94th percentile); Freckles (the 6th percentile); MC1R CC; MC1R GG; MC1R TT; SLC45A2 CG; TYR GG; Melanoma (24 significant genomic variants) (the 6th percentile); Basal cell carcinoma (the 20th percentile); Squamous cell carcinoma (the 3rd percentile); CDKN2A CC; MITF GG; ASIP GG; CASP8 CC	MC1R GG; IRF4 CC; SLC24A5 AA; HERC2 AG; OCA2 CC; Melanoma (505 variants, lassosum) (the 56th percentile); TERT CA; PLA2G6 CT
DNA Damage and DNA Repair Deficit	Number of nevi (signals) (the 88th percentile); Melanoma (previous report, proprietary method) (the 89th percentile); MTAP AA; PARP1 TT	Tanning difficulty (burning instead of tanning) (the 16th percentile); Solar burns in childhood (the 1st percentile); Skin color (darker) (the 94th percentile); Freckles (the 6th percentile); MC1R CC; MC1R GG; MC1R TT; SLC45A2 CG; TYR GG; Melanoma (24 significant genomic variants) (the 6th percentile); Basal cell carcinoma (the 20th percentile); Squamous cell carcinoma (the 3rd percentile); CDKN2A CC; MITF GG; ASIP GG; CASP8 CC	MC1R GG; IRF4 CC; SLC24A5 AA; HERC2 AG; OCA2 CC; Melanoma (505 variants, lassosum) (the 56th percentile); TERT CA; PLA2G6 CT
Genomic instability	Number of nevi (signals) (the 88th percentile); Melanoma (previous report, proprietary method) (the 89th percentile); MTAP AA; PARP1 TT	Melanoma (24 significant genomic variants) (the 6th percentile); Basal cell carcinoma (the 20th percentile); Squamous cell carcinoma (the 3rd percentile); CDKN2A CC; MITF GG; ASIP GG; CASP8 CC	Melanoma (505 variants, lassosum) (the 56th percentile); TERT CA; PLA2G6 CT; MX2 CT
Somatic / Tissue Mosaicism	Number of nevi (signals) (the 88th percentile); Melanoma (previous report, proprietary method) (the 89th percentile); MTAP AA; PARP1 TT	Melanoma (24 significant genomic variants) (the 6th percentile); Basal cell carcinoma (the 20th percentile); Squamous cell carcinoma (the 3rd percentile); CDKN2A CC; MITF GG; ASIP GG; CASP8 CC	Melanoma (505 variants, lassosum) (the 56th percentile); TERT CA; PLA2G6 CT; MX2 CT
Chronic Inflammation	FLG CG; CARD9 TT; TLR2 CA	FLG GG; Seborrheic dermatitis (the 9th percentile); CLEC7A AA; TLR2 GG	Acne (the 40th percentile); WNT10A TT; FST CT; SEMA4B CC
Dysbiosis	CARD9 TT; TLR2 CA	Seborrheic dermatitis (the 9th percentile); CLEC7A AA; TLR2 GG	Acne (the 40th percentile); WNT10A TT; FST CT; SEMA4B CC
Immunosenescence	CARD9 TT; TLR2 CA	Seborrheic dermatitis (the 9th percentile); CLEC7A AA; TLR2 GG	—
Intestinal Barrier dysfunction	FLG CG	FLG GG	—
Remodelling of the extracellular matrix	FLG CG	FLG GG; Facial wrinkles (the 0th percentile); Skin aging (microtopography) (the 5th percentile); Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	Facial aging ("appears older than it is") (the 29th percentile); Seborrheic keratoses (the 40th percentile); BNC2 AA; COL1A1 CC; NQO1 GG; AGER CC
Accumulation of Advanced Glycation Final Products (AGEs)	—	Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	COL1A1 CC; NQO1 GG; AGER CC
Redox imbalance	—	Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	COL1A1 CC; NQO1 GG; AGER CC
Endocrine dysregulation	—	—	Acne (the 40th percentile); WNT10A TT; FST CT; SEMA4B CC
Fragmentation of Elastin	—	Facial wrinkles (the 0th percentile); Skin aging (microtopography) (the 5th percentile)	Facial aging ("appears older than it is") (the

29th percentile);
Seborrheic keratoses
(the 40th percentile);
BNC2 AA

Facial aging ("appears
older than it is") (the
29th percentile);
Seborrheic keratoses
(the 40th percentile);
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MC1R GG; IRF4 CC;
SLC24A5 AA; HERC2 AG;
OCA2 CC; Facial aging
("appears older than it
is") (the 29th
percentile); Seborrheic
keratoses (the 40th
percentile); BNC2 AA

Cell Senescence	—	Facial wrinkles (the 0th percentile); Skin aging (microtopography) (the 5th percentile)
Oxidative stress	—	Tanning difficulty (burning instead of tanning) (the 16th percentile); Solar burns in childhood (the 1st percentile); Skin color (darker) (the 94th percentile); Freckles (the 6th percentile); MC1R CC; MC1R GG; MC1R TT; SLC45A2 CG; TYR GG; Facial wrinkles (the 0th percentile); Skin aging (microtopography) (the 5th percentile); Genetic glyated hemoglobin (HbA1c) (the 7th percentile)

Practical part

Routine and topical assets that genetics favors, crossed with the genetic plan v2 and current medication (oral isotretinoin 40 mg/week, rapamycin, lithium, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN, melatonin). The current stack remains as it is; everything that follows is suggestion (not applied). Base-routine: morning — soft cleaning, vitamin C, moisturizer with ceramides, sunscreen FPS 50; night — smooth cleaning, azelaic acid or peptides, moisturizing with ceramides; 2–3×/week — ketoconazole in the dermatitis zones.

1. Daily sunscreen FPS 50 wide spectrum

Genetics: Favourable sun, but melanoma: divergent signals (p7/p56 vs p89 anterior); nevi p88; MTAP AA; isotretinoin FPS 50 with high UVA protection (UVA in the circle or high PPD), every day, on the face, neck, ears and hands; reapply each 2 h outdoors. Sunglasses and hat. It is the anti-aging agent with better evidence (less wrinkles and less melanoma in trials). Isotretinoin increases sensitivity to the sun.

V2 genetic plan and medication: Similar to the v2 plan (item 14: daily photoprotection + annual dermoscopy). • *suggestion (not applied)*

2. Annual dermoscopy of the whole skin

Genetics: P88 nevi (3 loci); MTAP AA; PARP1 TT; previous report of melanoma p89

Annual examination with digital dermoscopy (signal mapping). Monthly self-examination (ABCDE rule).

V2 genetic plan and medication: Similar to the v2 plan. • *suggestion (not applied)*

3. Barriers: ceramides and soft cleaning

Genetics: FLG: rare absences; possible rs542799026 (unconfirmed); isotretinoin (dried)

Hydrating with ceramides, cholesterol and fatty acids (proportion close to 3:1:1) 2×/day; cleaning without sulfates, warm water, acid pH. Pantenol and glycerin help. Lip balm with vaseline or ceramides (dry lip of isotretinoin).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Topical vitamin C in the morning

Genetics: NQO1 and GPX1 typical; skin antioxidants; HbA1c p7

L-ascorbic acid 10–15% with vitamin E and ferulic acid (folds the photoprotection of the protector). If burning with isotretinoin, use a smoother form (ascorbyl glucoside or tetrahexyldecyl ascorbate).

V2 genetic plan and medication: Compatible; oral vitamin C is already in the plan (S16). • *suggestion (not applied)*

5. Topical retinoids: only after isotretinoin

Genetics: Collagen (COL1A1 without "s"); p0.3 wrinkles

Topical tretinoin (0.025–0.05% at night) is the active with better evidence for photoaging, but it is added to oral isotretinoin (irritation, dryness, desquamation). While isotretinoin remains: no topical retinoids, except for dermatologist indication. After stopping: start with 2–3 nights per week.

V2 genetic plan and medication: Coherent with the v2 plan (zero oral retinol; isotretinoin × retinoids). • *suggestion (not applied)*

6. Azelaic acid

Genetics: Acne p40; seborrheic dermatitis; pigmentation

15–20% at night: anti-inflammatory, slightly antifungal and antibacterial stains and is compatible with isotretinoin (with moisturizer). Good for areas of seborrheic dermatitis and redness.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

7. Ketoconazole for seborrheic dermatitis

Genetics: CARD9 TT; TLR2 P631H (weak); Seborrheic dermatitis score p9

Ketoconazole 2% in shampoo or cream 2–3×/week in the affected areas (hair, eyebrows, nasogenic grooves, barb), allowing to act 3–5 min; alternate with ciclopirox 1% or zinc pyrithione. Maintenance 1×/week when controlled. Topical ketoconazole does not interact with rapamycin (despicable absorption).

V2 genetic plan and medication: Similar to the v2 plan (topic antifungals; oral fluconazole only outside the 2 weeks after rapamycin administration). • *suggestion (not applied)*

8. Complementary peptides and antioxidants

Genetics: P0.3 wrinkles; p5 skin aging (little loci)

Signaling peptides (palmitoyl pentapeptide-4, copper tripeptide GHK-Cu) at night as an alternative to retinoids during isotretinoin; moderate to weak evidence. Topical green tea or topical resveratrol are optional.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

9. Topical Rapamycin (experimental)

Genetics: Skin senescence (mechanism, non-gene)

In a small trial, topical rapamycin in low concentration reduced senescence markers (p16) and improved skin collagen. Experimental; only with the doctor, who already prescribes oral rapamycin.

V2 genetic plan and medication: New; weak evidence. • *suggestion (not applied)*

10. Photoprotection and antiglycation

Genetics: HbA1c p7; AGER CC

Maintain astaxanthin 12 mg, lutein/zeaxanthin, omega-3, collagen 20 g, carnosine and vitamin C, already in the stack. Polypodium leucotomos extract (240–480 mg before sun exposure) has moderate evidence of oral photoprotection.

V2 genetic plan and medication: Astaxanthin (S22), lutein (S24), carnosine (S21), collagen (S9) already in the plan; Polypodium would be new. • *suggestion (not applied)*

11. Procedures during isotretinoin

Genetics: Barriers; healing (see report Nails, Wound healing and Skin Inflammation)

Avoid ablative laser, dermoabrasion, medium/deep peeling and waxy waxing during isotretinoin and in the following months, except for dermatologist's decision.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

Avoiding or with caution

- Soils and intense sun exposure without protection (isotretinoin; nevi p88). *suggestion (not applied)*
- Products with topical retinol or retinoids while taking oral isotretinoin. *suggestion (not applied)*
- Aggressive mechanical spoliants, diaries strong acids, denatured alcohol and fragrances in the routine (more fragile barrier with isotretinoin). *suggestion (not applied)*
- Pure vegetable oils in areas of seborrheic dermatitis (feeding Malassezia); prefer moisturizers without oil or with squalane. *suggestion (not applied)*
- Smoking and passive smoking (aging of the skin; PiMZ). *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. 2282del4 deletion of FLG and the very rare variants of MC1R and FLG are almost never read or imputed with confidence.
- Rare variants read in the chip (e.g., FLG nonsense) can be false positives; only sequencing confirms.
- The polygenic scores of melanoma and other skin cancers explain a small part of the risk; the main factor is accumulated sun exposure (and burns), the number of nevus and family history.
- The scores of wrinkles, skin aging and seborrheic dermatitis have few loci; low reliability.
- The skin responds much more to photoprotection, routine and medication (isotretinoin) than to genes.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001937 (PGS Catalog; Penalized regression (bigstatsr); Ease of skin tanning; PGP000263, PMID 34995502 · PGS001257 (PGS Catalog; snpnet; Childhood sunburn; PGP000244, PMID 35324888 · PGS001897 (PGS Catalog; Penalized regression (bigstatsr); Skin color; PGP000263, PMID 34995502 · PGS000079 (PGS Catalog; Genome-wide signifiers; Melanoma; PGP000050, PMID 33579919 · PGS000415 (PGS Catalog; lassosum; Melanomas of skin (diagnosis or history); PGP000118, PMID 32991828 · PGS000448 (PGS Catalog; lassosum; Basal cell carcinoma; PGP000118, PMID 32991828 · PGS000464 (PGS Catalog; Pruning and Thresholding (P+T); Squamous cell carcinoma; PGP000118, PMID 32991828 · PGS001072 (PGS Catalog; snpnet; Facial aging, looking 'older than you are'; PGP000244, PMID 35324888 · PGS001140 (PGS Catalog; snpnet; Seborrheic keratosis (time-to-event); PGP000244, PMID 35324888 · PGS001953 (PGS Catalog; Penalized regression (bigstatsr); Glycated haemoglobin (HbA1c); PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

Generated in 02-10-2026 (local time) by the scripts of this report.